

Today

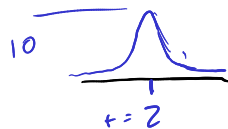
Adaptive Runge Kutta
Stiff ODEs
Implicit Methods

Announcements

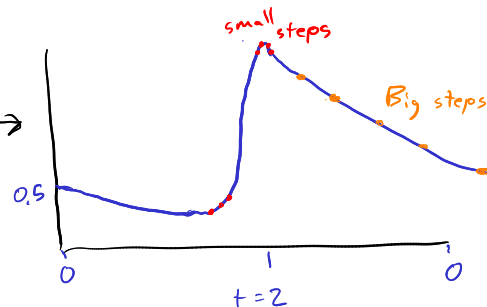
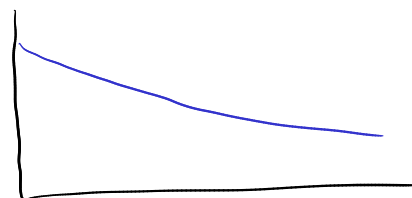
Lab 2, HW4 due tomorrow

$$\dot{x} = -0.6x \Rightarrow x = x_0 e^{-0.6t}$$

$$\dot{x} = 10 e^{-(t-2)^2 / [2(0.075)^2]} - 0.6x$$



takes value of 10 at $t=2$



Option 1

Step Halving

Option 2: Adaptive Runge Kutta

```

1: function EULER_ADAPTIVE(f, [t0, tf], x0, Es)
2:   t ← [t0], x ← [x0], h ← (tf - t0)/100
3:   while tend < tf do
4:     h ← min(h, tf - tend)
5:     xfull ← xend + h f(tend, xend)
6:     xmid ← xend + h/2 f(tend, xend)
7:     xhalf ← xmid + h/2 f(tend + h/2, xmid)
8:     Ea ← |xhalf - xfull|
9:     if Ea ≤ Es then
10:      append tend + h to t
11:      append xhalf to x
12:     end if
13:     h ← 0.9 h √(Es/Ea)
14:   end while
15:   return t, x
16: end function
  
```

full Euler's
2 Euler steps with $h/2$
▷ or RK4
▷ b/c E_a is $O(h^2)$

k_i for next iter

```

1: function BS23_ADAPTIVE(f, [t0, tf], x0, Es)
2:   t ← [t0], x ← [x0], h ← (tf - t0)/100
3:   k1 ← f(t0, x0)
4:   while tend < tf do
5:     h ← min(h, tf - tend)
6:     k2 ← f(tend + h/2, xend + h/2 k1)
7:     k3 ← f(tend + 3h/4, xend + 3h/4 k2)
8:     x3 ← xend + h (2/9 k1 + 1/3 k2 + 4/9 k3)
9:     k4 ← f(tend + h, x3)
10:    x2 ← xend + h (7/24 k1 + 1/4 k2 + 1/3 k3 + 1/8 k4)
11:    Ea ← |x3 - x2|
12:    if Ea ≤ Es then
13:      append tend + h to t
14:      append x3 to x
15:      k1 ← k4
16:    end if
17:    h ← 0.9 h (Es/Ea)^(1/3)
18:  end while
19:  return t, x
20: end function
  
```

3 function evaluations per step
▷ 3rd order
▷ 2nd order
▷ slope at the new x_{end}
▷ b/c E_a is $O(h^3)$
Local: $O(h^3)$

ode23

ode45 does the same thing with 4th/5th order

Stiff ODEs

A "stiff" system involves rapidly-changing and slowly-changing components

Van der-pol Oscillator

$$\dot{x} - \mu(1-x^2)\dot{x} + x = 0$$

first create system $\dot{\vec{x}} = f(t, \vec{x})$

$$v \equiv \dot{x}$$

$$\dot{\vec{x}} = f(t, \vec{x})$$

$$\begin{bmatrix} \dot{v} \\ \dot{x} \end{bmatrix} = \begin{bmatrix} \mu(1-x^2)v - x \\ v \end{bmatrix}$$

system of 1st order ODEs



$\mu=1$: not stiff

RK4 works

BS23 works

$\mu=1000$: stiff

RK4 diverges

BS23 takes a lot of steps

Implicit methods

Help stability not accuracy

Implicit Euler Update

Explicit $x_{i+1} \leftarrow x_i + f(t_i, x_i)h$

Implicit $x_{i+1} \leftarrow x_i + f(t_i + h, x_{i+1})h$

How do we find x_{i+1} ?

Solve (possibly nonlinear equation)

$$x_{i+1} - x_i - f(t_{i+1}, x_{i+1})h = 0$$

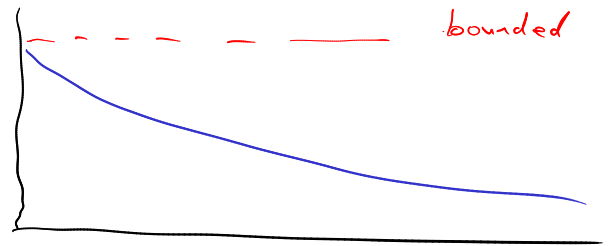
Use any nonlinear solver

Example

$$\dot{x} = -ax \quad a > 0$$

unstable if errors grow exponentially
for a bounded solution

Explicit Euler was conditionally stable



Implicit: $x_{i+1} = x_i - ax_{i+1}h$

$$x_{i+1}(1+ah) = x_i$$

$$x_{i+1} = \frac{x_i}{1+ah}$$

← shrinks
regardless
of h

Implicit Euler is
unconditionally stable

ode23s in Matlab

↑ stiff